

PROJECT UPDATE 6

Final Integration, Bidirectional AI Firewall, and Demonstration Readiness

AI-Powered E-Commerce Platform with Intelligent Business Insights

SUMMARY - Update 6 completes the final integrated build of the project. Shakib's React/Vite storefront, Rahat's Django REST/PostgreSQL backend, and Farhan's dual Ollama chatbot layer now operate together in one system. The AI layer was further strengthened with a bidirectional firewall that filters both incoming prompts and generated responses, with verified chatbot rate limiting.

1. FRONTEND FINALIZATION - SHAKIB

The final frontend remains a React 19 single-page application built with Vite 8, Tailwind CSS 4 and React Router. Shakib completed the customer-facing shopping flow and the interfaces used to communicate with the Django APIs.

- Route coverage:** home, login, registration, products, product details, categories, cart, checkout, orders, profile, customer AI, admin AI and fallback page.
- API connection:** store pages communicate with the integrated REST backend, while protected operations use JWT Bearer authentication from the client.
- AI interfaces:** customer and administrator chat pages send requests to the final /api/ai/chat/ and /api/ai/admin-chat/ endpoints and display backend errors and responses.
- Final usability:** responsive layouts, loading states, navigation, empty states and error handling support the demonstration flow.

2. BACKEND AND POSTGRESQL - RAHAT

Rahat's backend is now present in the final integrated build using Django and Django REST Framework with PostgreSQL configured as the primary database engine.

- Backend modules:** accounts, store, cart, orders, chat, core and related services expose the project's main REST resources.
- Database integration:** the final settings use the PostgreSQL backend with environment-configurable database name, user, password, host and port.
- Authentication:** JWT authentication and refresh endpoints remain integrated for protected frontend requests and administrator operations.
- Business APIs:** product/category, inventory, cart, order, payment, invoice, profile and chat-related routes are available through the combined backend.

3. FINAL INTEGRATION STATUS

- Integrated stack:** React/Vite frontend -> Django REST APIs -> PostgreSQL database, with local Ollama **phi3:mini** serving the customer and administrator AI endpoints.
- Customer AI:** uses a sanitized public product snapshot containing product name, category, price, description and Available/Out of stock status.
- Admin AI:** requires Django administrator permission and receives a read-only database snapshot for products, stock, customers, orders and sales summaries.

4. AI FIREWALL AND CHATBOTS - FARHAN

Farhan completed the dual Ollama chatbot integration and added a lightweight bidirectional AI firewall around both chatbot endpoints. The security layer operates before and after model generation instead of relying only on system-prompt instructions.

- Input normalization:** Unicode text is normalized, zero-width/control characters are removed, and repeated whitespace is collapsed before security checks and model processing.
- Prompt filtering:** prompt-injection, hidden-prompt extraction, security-bypass and protected-data requests are rejected before Ollama is called.
- Output filtering:** generated replies are checked for secret-like assignments, authentication material, database credentials and protected prompt leakage before being returned.
- JWT leak detection:** JWT-shaped tokens and Bearer-token output are specifically detected and blocked by the output firewall.
- Rate limiting:** Django REST Framework ScopedRateThrottle limits the customer AI to 10 requests/minute and the admin AI to 20 requests/minute.

5. VERIFICATION RESULTS

DJANGO CHECK	No issues detected
NORMAL AI	Customer/admin replies work
INPUT FIREWALL	Injection request blocked
OUTPUT FIREWALL	Secret/JWT tests blocked
RATE LIMIT	11th customer request -> 429

6. OUTCOME AND NEXT ACTIONS

- Update 6** produced a single demonstrable system in which the frontend, backend, PostgreSQL database and both local AI assistants are integrated. The new firewall adds clear input-side and output-side AI security controls while preserving normal customer and administrator chatbot use.
- Repository:** push the final verified integrated files and this Update 6 report to the group GitHub repository.
- Demonstration:** prepare the one-minute screen recording and final demonstration material using this working build.
- Final check:** repeat the core customer flow and both chatbot security tests before submission/presentation.

EVIDENCE REVIEWED: final Masterfinal ZIP; React/Django source files; PostgreSQL configuration; dual Ollama chatbot routes; successful `python manage.py check`; working customer/admin chatbot tests; input/output firewall tests; verified HTTP 429 throttling on the 11th customer request.